

Overview

This exercise applies group recommender system (GRS) techniques to a movie dataset. Two groups are selected and their members' individual ratings are aggregated using three strategies; Average, Least Misery, and Most Pleasure in order to generate a ranked list of recommendations for unseen movies. The predicted ratings are computed via cosine similarity between genre vectors, and the results are compared with the actual group decisions recorded in the dataset.

1 Selected Groups

Two groups were selected from the provided dataset and analysed using three aggregation strategies each.

Table 1: Selected groups and their member user IDs.

Group	User IDs
Group 3	U11, U15, U29
Group 7	U1, U6, U29

2 Methodology

The recommendation pipeline follows a content-based / collaborative-filtering hybrid approach in four steps:

Step 1. Aggregate individual ratings. For each of the 50 rated movies, apply the aggregation strategy across group members' filled ratings (missing entries replaced by the movie's global average).

Step 2. Compute cosine similarity. For every pair (new movie, rated movie), compare their 18-dimensional binary genre vectors:

$$\text{sim}(A, B) = \frac{\mathbf{v}_A \cdot \mathbf{v}_B}{\|\mathbf{v}_A\| \|\mathbf{v}_B\|}$$

Step 3. Predict group rating for each new movie.

$$\hat{r}_G(\text{new}) = \max_{\text{rated}} (r_G(\text{rated}) \times \text{sim}(\text{new}, \text{rated}))$$

where $r_G(\text{rated})$ is the aggregated group rating of the rated movie.

Step 4. Rank and recommend. New movies are sorted by predicted score; the top 3 form the final recommendation list.

Aggregation Strategies

Three strategies from the literature (Masthoff, 2004) were applied:

- **Average** — arithmetic mean of the group members' ratings: $r_G^{\text{avg}}(i) = \frac{1}{|G|} \sum_{u \in G} r_u(i)$
- **Least Misery** — minimum rating across members; ensures no member is severely dissatisfied: $r_G^{\text{LM}}(i) = \min_{u \in G} r_u(i)$
- **Most Pleasure** — maximum rating across members; maximises peak satisfaction: $r_G^{\text{MP}}(i) = \max_{u \in G} r_u(i)$

3 Results

Top-3 Recommendations – all groups and strategies:

		1st	2nd	3rd
Group	Strategy			
Group 3	Average	HellBoy II: The Golden Army	Incredible Hulk, The	Quantum of Solace
	Least Misery	HellBoy II: The Golden Army	Incredible Hulk, The	Quantum of Solace
	Most Pleasure	Forgetting Sarah Marshall	Sex and the City	HellBoy II: The Golden Army
Group 7	Average	Forgetting Sarah Marshall	Sex and the City	Burn After Reading
	Least Misery	Burn After Reading	Forgetting Sarah Marshall	Sex and the City
	Most Pleasure	Forgetting Sarah Marshall	Sex and the City	Quantum of Solace

Figure 1: Top-3 recommendations per group and aggregation strategy

4 Analysis of Differences Between Strategies

Group 3 (U11, U15, U29)

Average and **Least Misery** produce identical top-3 lists (*HellBoy II*, *Incredible Hulk*, *Quantum of Solace*), all action/fantasy films. This coincidence arises because the three members' ratings for action movies are relatively close to each other so even the minimum remains high enough to outrank comedy titles.

Most Pleasure promotes *Forgetting Sarah Marshall* and *Sex and the City* to the top. At least one member rates those comedies very highly and the strategy amplifies that individual preference regardless of how the other members feel.

Group 7 (U1, U6, U29)

Average and **Most Pleasure** agree on the top 2 (*Forgetting Sarah Marshall*, *Sex and the City*). **Least Misery** diverges by placing *Burn After Reading* first. While two members rate the comedies highly, the third member's lower rating pulls them down under the minimum operator, whereas *Burn After Reading* rated more consistently across all three rises to first place.

Which strategy is preferable?

There is no universally best strategy; the choice depends on context:

- **Least Misery** is most equitable for diverse groups and avoid recommendations that would clearly dissatisfy any member.
- **Average** provides a balanced compromise and is robust against outliers.
- **Most Pleasure** risks favouring one enthusiastic member while leaving others indifferent.

5 Comparison with Real Experimental Results

Actual group decisions (Table 3):

	Actual 1st	Actual 2nd	Actual 3rd
Group			
Group 3	Indiana Jones...Crystal Skull	Kung Fu Panda	Sex and the City
Group 7	Be Kind Rewind	Burn After Reading	10,000 B.C.

Figure 2: Actual group decisions recorded in the dataset (Table 3)

The computed recommendations differ considerably from the actual group decisions:

- **Group 3** actually chose *Indiana Jones* and *Kung Fu Panda* as their top 2, neither appeared in any computed top-3. *Sex and the City* (M11), which appeared only in the Most Pleasure ranking, was selected third in reality.
- **Group 7** chose *Be Kind Rewind* (M13) first, absent from all computed rankings and *10,000 B.C.* (M8) third, also missing from our results. Only *Burn After Reading* (M5) matches, appearing first under Least Misery.

Reasons for divergence:

- **Social dynamics** Real groups negotiate and compromise; algorithmic strategies treat each member independently.
- **Average imputation** Missing ratings are filled with the movie’s global average which can distort a member’s apparent preferences.
- **Coarse genre vectors** The binary genre representation is a rough approximation; two movies sharing genres may be perceived very differently.
- **Context and novelty** Real users consider whether they have already seen a film, their current mood, novelty and none of which are modelled here.

6 Advantages and Disadvantages of the Strategies

Table 2: Comparison of the three aggregation strategies.

Strategy	Advantages	Disadvantages
Average	Fair to all members; intuitive; robust against single outliers.	May not represent any one member’s actual preference; sensitive to rating-scale differences.
Least Misery	Ensures no member is strongly dissatisfied; good for diverse groups; safe and conservative.	One low-rating member can veto popular items; may produce uninspiring recommendations.
Most Pleasure	Maximises peak satisfaction; useful when one member has clear authority or expertise.	Ignores dissatisfied members; easily dominated by a single enthusiastic user.

7 Conclusion

This exercise showed that choosing a different aggregation strategy can lead to very different recommendations for the same group. There is no single best strategy and it depends on the group situation: Least Misery works well when members have very different tastes, Average is a safe and fair choice in most cases, and Most Pleasure makes sense when one member’s preference naturally guides the group. Comparing our results with actual group decisions also shows that real people do not always behave the way a mathematical model predicts, which is why social and contextual factors matter in practice.